Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [14EE13/23]

FIRST SEMESTER B.TECH EXAMINATION, DECEMBER-2015

BASICS OF ELECTRICAL ENGINEERING

Time:3 Hours **Maximum Marks:70**

Instructions : Answer five full questions, choosing at least TWO from each part.

PART - A

1. a. State Kirchhoff's latus. **04**

   b. An 8Ω resistor is in series with a parallel combination of two resistors 12Ω and 6Ω. If the current in the 6Ω resistor is 5A, determine the total power dissipated in the circuit. **05**

   c. Determine the current drawn from the source in the network shown in Fig. 1c. **05**

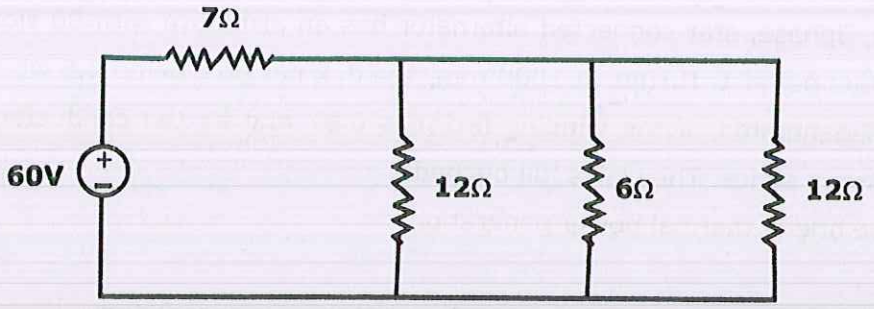

Fig. 1c

2. a. Define from factor of an alternating quantity. Derive the expression for it. **08**

   b. A resistance of 20. An inductance of 0.2H and a capacitance of 100 μF are connected in series across a 220V, 50Hz supply. Calculate the current, power factor & power consumed in the circuit. **06**

3. a. With the aid of a phasor diagram obtain the relationship between the line and phase voltages in a three- phase, star connected system. **07**

   b. Write short note on: **07**

      i. Fuse

      ii. Precautions against shock

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4. a. Derive the emf equation for a dc generator. **04**

   b. Explain the principle of operation of dc motor. **04**

   c. A 4pole, lap connected dc generator runs at 900 rpm and has 40 slots, each slot containing 10 conductors. **06**

      i. find the useful flex per pole, if the emf generated is 300V.

      ii. what is the average value of the emf induced in a field coil having 1000 turns, when the field coil is opened and the flux becomes zero in 0.05 sec.

PART - B

5. a. Derive the EMF equation of single phase transformer. **06**

b. A 50 KVA, 3300/330V, 1 ø transformer has iron loss and full load copper loss of 400 watts and 600 watts respectively. Calculate the efficiency at half full load and 0.9 Pf. Also calculate the load at which the maximum efficiency occurs. **04**

c. A3ø, 6pole,50Hz induction motor has a slip of 1% at no load and 3% at full load. Find, **04**

i. synchronous speed

ii. the no load speed

iii. full load speed

6. a. What are the advantages of rotating field synchronous generator? **04**

b. A 6pole, 3phase, star connected alternator has an armature with 90 slots and 12 conductors per slot. It runs at 1000 rpm, the flux per pole being 0.5 wb. Calculate the emf generated, if the winding factor is 0.97 and all the conductors in each phase are in series. The coil is full pitched. **06**

c. Describe briefly thermal power generation. **04**

7. a. List out the necessary requirements of a good distribution system.

b. Explain various aspects of energy conservation.

c. Explain the necessity of energy storage. **06**

8. a. Mention the limitations of tidal energy. **08**

b. Write a short note on:

i. Wind energy

ii. Biomass energy